

Extended Abstract

Motivation Inference is becoming the dominant cost of deployed language models, and agentic workloads make that cost increasingly significant. A single agent task can include long reasoning traces, tool calls, retries, file operations, and environment interaction. Its energy is therefore not determined by one forward pass or by model size alone. We examine intelligence per joule (IPJ) as a first-class objective for agent inference and study two ways to optimize it: selecting the right harness for the task and post-training the model against measured energy rather than token length.

Method We measure IPJ and intelligence per watt with per-query NVML telemetry, attributing GPU energy and power to completed agent trajectories. We treat the harness as part of the optimization problem because it sets the context, tool-use loop, and retry behavior seen by the model. We then compare two post-training objectives on the same task split: the token-oriented O1-Pruner baseline and an energy-aware variant that replaces response length with measured joules over the full trace.

Implementation Our measurement study covers 2 models, 5 harnesses, and 13 benchmarks, yielding 67 validated model-task-harness configurations. The harnesses range from lightweight reasoning calls to multi-turn tool and terminal agents. For post-training, we fine-tune Qwen3.5-27B on SuperGPQA with a fixed 80/20 split, $K = 12$ reference samples per question, the same one-epoch training recipe for both methods, and a fixed GPT grader held constant across base and post-trained evaluations.

Results Harness choice is the largest immediate lever we observe. For a fixed model and benchmark, the best and worst harness differ by up to $54\times$ in IPJ, with a median $3\times$ gap. On SuperGPQA and MMLU-Pro, lighter scaffolds match heavier ones within a few accuracy points at 4 to $9\times$ lower energy. Output tokens explain 94% of the variance in measured energy across scorable cells, but token reduction alone does not solve agent efficiency. O1-Pruner lowers accuracy on all four SuperGPQA harnesses by 6 to 21 points. The energy-aware variant improves the post-training tradeoff, has higher accuracy than O1-Pruner on all four harnesses, and reaches $2.51\times$ FP8-equivalent relative IPJ on Terminus.

Discussion To our knowledge, this is the first work to motivate task-specific harness selection as a significant lever for agent energy efficiency. The result is practical because it does not require new weights: efficiency comes from choosing whether a task needs a heavy agent loop or a lighter scaffold. The post-training results show the complementary point. Tokens are correlated with energy in aggregate, but they are a lossy proxy once the model is embedded in an agent loop. An energy- and agent-aware reward is better matched to deployment because it scores the complete trajectory.

Conclusion Agent inference should be optimized at the trajectory level. The strongest path is to combine the two levers studied here: route each task to an appropriate harness, then train or select policies using energy measured on that harness. This makes IPJ a deployment objective rather than an after-the-fact diagnostic.

Intelligence per Joule as a First-Class Post-Training Objective

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Abstract

We propose intelligence per joule (IPJ) as a first-class objective for agent inference and study two levers for optimizing it: task-specific harness selection and energy-aware post-training. Agent tasks include reasoning, tool calls, retries, and environment interaction, so their cost is determined by the full trajectory rather than a single forward pass. We measure per-query GPU energy with NVML across 2 models, 5 harnesses, and 13 benchmarks, totaling 67 validated configurations. Our results demonstrate that selecting the correct harness for a task is an immediate efficiency lever. To our knowledge, this is the first work to motivate task-specific harness selection as a significant lever for agent energy efficiency: for a fixed model and benchmark, the best and worst harness differ by up to $54\times$ in IPJ, with a median $3\times$ gap, and lighter scaffolds match heavier ones within a few accuracy points at 4 to $9\times$ lower energy on SuperGPQA and MMLU-Pro. We then show that naive token-only post-training is not sufficient for agent efficiency: O1-Pruner lowers accuracy across harnesses because token length is a lossy proxy once the model is embedded in an agent loop. In contrast, our energy- and agent-aware variant optimizes measured joules over complete traces, improves the post-training tradeoff, and reaches $2.51\times$ FP8-equivalent relative IPJ on Terminus. These results support optimizing deployment harnesses and model policies together, using measured trajectory energy rather than tokens alone.

1 Introduction

Inference, not training, is becoming the dominant cost of deploying large language models, as a model is trained once but serves a growing volume of queries over its lifetime; inference is projected to account for roughly two-thirds of AI compute in 2026, up from one-third in 2023 and half in 2025 (Stewart et al., 2025). This pressure is amplified by the shift toward long-horizon agentic workloads, where a single task spans a long trajectory of reasoning, tool calls, retries, and environment interaction rather than a single forward pass. The share of tokens routed to reasoning-optimized models rose from a negligible slice in early 2025 to over half by the end of the year (OpenRouter, 2025). The system-level effect is large. Oviedo et al. (2026) project that at one billion queries per day, raising the share of long-reasoning traffic from 0 to 10% increases serving energy from 0.81 to 1.80 GWh per day, a 220% increase (Figure 1).

Prior work measures inference energy and standardizes it into the hardware-agnostic intelligence-per-watt (IPW) and intelligence-per-joule (IPJ) metrics (Samsi et al., 2023; Luccioni et al., 2024; Saad-Falcon et al., 2026), for off-the-shelf models under fixed inference configurations. Other work, (Han et al., 2025; Hou et al., 2025; Aggarwal and Welleck, 2025; Luo et al., 2025; He et al., 2026)) takes token count as a proxy for cost and successfully apply post-training techniques to reduce model output tokens. However, token count is a lossy surrogate for energy (Kaiser et al., 2026).

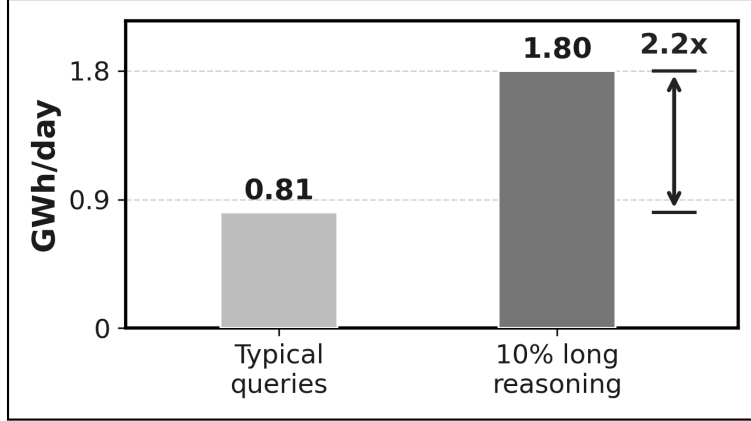


Figure 1: Projected serving energy as long-reasoning traffic grows Oviedo et al. (2026).

Thus, we treat measured energy as a first-class objective and study two levers that control it: the agent harness, which sets the context and tool-use pattern at inference time, and the model policy, as set by post-training. This framing motivates three research questions: (i) how effective is task-dependent harness selection at improving energy efficiency (ii) do token- or length-oriented post-training methods improve agent energy efficiency, and (iii) what reward should post-training optimize if the goal is joules rather than tokens.

We make three contributions, one per research question. First, we conduct a comprehensive measurement of agent inference energy across 2 models, 5 harnesses, and 13 benchmarks. Totaling 67 configurations at real inference scale, using per-query NVML telemetry. We then present the first evaluation of a token-efficiency post-training method under measured energy, applying O1-Pruner (Luo et al., 2025) to Qwen3.5-27B (Qwen Team, 2026) on SuperGPQA (Team et al., 2025). Finally, we introduce an energy-harmonizing reward that optimizes accuracy per joule rather than accuracy per token and evaluate it against token-based post-training.

2 Related Work

Samsi et al. (2023) established empirical baselines for inference energy, finding roughly three to four joules per output token for LLaMA-65B on V100/A100 GPUs, and noting that inference energy receives far less attention than training cost. Luccioni et al. (2024) extended these measurements across a broader range of models and tasks. Saad-Falcon et al. (2026) introduce IPW and IPJ as unified cross-hardware efficiency metrics, conducting the first large-scale study across more than one million queries, 20+ models, and eight hardware accelerators, and find that local accelerators achieve at least $1.4\times$ lower IPW than cloud accelerators running identical models. Oviedo et al. (2026) project the system-level energy of agentic serving and show that growing long-reasoning traffic sharply increases datacenter energy. Critically, all of these works study off-the-shelf models under fixed inference configurations, and none examine how post-training interventions, or the choice of agent harness, affect a model’s energy profile.

A growing number of papers address overthinking in chain-of-thought models, the tendency of RL-trained reasoning models to produce excessively long responses even for simple queries. Methods fall broadly into prompt-based, output-based, and training-based approaches. TALE (Han et al., 2025) exemplifies the prompt-based approach, reducing token cost by dynamically estimating and injecting a token budget at inference time. Training-based RL approaches go further: ThinkPrune (Hou et al., 2025) trains with iteratively tightening hard token caps; L1/LCPO (Aggarwal and Welleck, 2025) trains via GRPO to satisfy prompt-specified budgets; O1-Pruner (Luo et al., 2025) applies RL-style fine-tuning under an accuracy constraint; and IAPO (He et al., 2026) uses information-aware advantage shaping to penalize low-importance tokens. All measure success in tokens, never in energy.

Kaiser et al. (2026) show that even within token-based efficiency, verbalization overhead varies $9\times$ across models and efficiency rankings diverge substantially from accuracy rankings (Spearman $\rho = 0.63$). This suggests that different RL methods, which cut tokens via qualitatively different mechanisms, may yield different energy profiles even at matched token counts. None of this work

measures hardware-level energy, leaving open how effectively token reduction translates into lower energy use.

Agent scaffolds such as ReAct, OpenHands, and Terminus wrap a base model in differing context-construction and tool-use loops, so the same model can consume very different input contexts and budgets on the same task. Because agent inference cost is trajectory-level, accumulating repeated context reprocessing and per-tool overhead rather than a single forward pass, it is not captured by single-pass energy measurements. This motivates treating the harness as a first-class variable in any study of agent energy.

3 Method

We quantify efficiency with two hardware-grounded metrics from Saad-Falcon et al. (2026), intelligence per joule (IPJ) and intelligence per watt (IPW),

$$\text{IPJ} = \frac{\text{accuracy}}{E_{\text{GPU}}(\text{J})}, \quad \text{IPW} = \frac{\text{accuracy}}{\bar{P}_{\text{GPU}}(\text{W})},$$

where accuracy is the fraction of scorable queries answered correctly (resolved over completed minus unscorable), E_{GPU} is total GPU energy, and $\bar{P}_{\text{GPU}}$ is mean GPU power. We instrument every run with a profiling harness that samples NVML at 50 ms and attributes energy and power to each query and each tool call. Per query, we log GPU energy and power, input and output tokens, turns, tool calls, wall-clock time, and correctness. Correctness is decided by a fixed language-model grader (gpt-5-nano) held constant across all base and post-trained comparisons, so accuracy differences reflect the model rather than the judge.

Energy has two levers. The harness fixes the context and tool-use loop at inference time, while the model policy can be changed by post-training. We study harness selection as a context-level intervention and compare two weight-level interventions, the token-based O1-Pruner below and the energy-aware objective of Section 3.2.

3.1 O1-Pruner

O1-Pruner (Luo et al., 2025) shortens reasoning with a length-harmonizing reward that compares each sampled response to a per-prompt reference and rewards being shorter while preserving accuracy,

$$R_{\text{LH}}(x, y) = \frac{\bar{L}_{\text{ref}}(x)}{L(y)} - 1 + \lambda(A(x, y) - \bar{A}_{\text{ref}}(x)),$$

where $L(y)$ is response length in tokens, $A(x, y)$ is binary correctness, and the reference length and accuracy are estimated from K presampled responses from the reference policy. The objective is optimized with a PPO-style clipped surrogate against the reference policy. We adapt it to SuperGPQA by replacing the math-equivalence reward with A-to-J multiple-choice correctness, and we keep the released training procedure unchanged so our results reflect the method as published.

3.2 Energy-Aware Reward

O1-Pruner treats response length as the efficiency target. We instead propose a Joule-harmonizing reward that replaces token length with measured inference energy, making energy a first-class post-training objective. For each prompt x , we compare a sampled response y against K reference-policy samples $y'_i \sim \pi_{\text{ref}}(\cdot | x)$:

$$R_{\text{JH}}(x, y) = \frac{\bar{J}_{\text{ref}}(x)}{J(y)} - 1 + \lambda(A(x, y) - \bar{A}_{\text{ref}}(x)),$$

where

$$\bar{J}_{\text{ref}}(x) = \frac{1}{K} \sum_{i=1}^K J(y'_i), \quad \bar{A}_{\text{ref}}(x) = \frac{1}{K} \sum_{i=1}^K A(x, y'_i).$$

Here $J(y)$ is the measured GPU energy in joules for the full inference trajectory, including decoding and agent-tool interaction, and $A(x, y)$ is binary task correctness. The ratio term rewards responses

that consume less energy than the prompt-specific reference baseline, while the accuracy term prevents the policy from reducing energy by sacrificing correctness.

This objective is structurally analogous to O1-Pruner’s length-harmonizing reward but targets hardware energy directly rather than token count. We optimize it with the same PPO-style clipped surrogate against the reference policy, replacing length measurements with per-query NVML energy measurements. This makes the reward sensitive to long-horizon agent behavior: a policy is rewarded not merely for producing fewer tokens, but for completing the full task trajectory with lower measured energy at comparable accuracy.

4 Experimental Setup

We evaluate Gemma-4-31B-it in bf16 and Qwen3.5-27B (Qwen Team, 2026) in FP8 across 13 benchmarks: WildChat, Arena-Hard-Auto v2, SuperGPQA, MMLU-Pro, TerminalBench, GAIA, FRAMES, SWE-bench Verified, LiveCodeBench, SWEfficiency, DeepResearchBench, LiveResearchBench, and BrowseComp. We report 67 validated model–task–harness cells, using only benchmark–harness pairings that match the task interface.

4.1 Harnesses and Evaluation Protocol

The harnesses are dspy-rlm, ReAct, OpenHands, Terminus, and ForgeCode. They span single-turn and structured reasoning calls as well as multi-turn workspace, shell, code, file, and tool-use workflows. TerminalBench uses the Terminus TerminalBench adapter.

Each benchmark uses a deterministic query order shared across models and harnesses. We use the official split when it has at most 500 examples; otherwise, we use a fixed 500-query subset. Smaller benchmarks keep their natural sizes, including 80 TerminalBench tasks, 165 GAIA tasks, 400 LiveCodeBench tasks, 104 SWEfficiency tasks, and 100-query research benchmarks.

All accepted local runs use one NVIDIA H100 per run, with the model server, evaluator, and energy monitor pinned to the same GPU. We track accuracy, completion and timeout state, unscorable reasons, wall-clock latency, input, output, and total tokens, throughput, turns, tool-call counts, tool names and latencies, GPU energy and power, CPU energy and power when available, MBU average and maximum when available, energy per output token, IPJ, IPW, and normalized summary statistics. Multi-turn runs additionally record these token, tool, latency, and energy measurements at the turn level.

4.2 O1-Pruner Post-Training

For post-training, we fine-tune Qwen3.5-27B on SuperGPQA (Team et al., 2025) with a fixed 80/20 split: seed 224, keyed by question UUID, with 21,223 train and 5,306 test questions. This ensures that every method trains and evaluates on the same questions. We presample $K = 12$ references per question, build the length-harmonizing targets with $\alpha = 5$, and fine-tune the last 48 transformer layers for one epoch at learning rate 2×10^{-6} with an effective batch size of 32 and a 10,000-token cutoff. We then evaluate the base and pruned models on the held-out test questions across all four single- and multi-turn harnesses under the fixed grader.

4.3 Energy-Aware O1 Variant

For the energy-aware O1 variant, we keep the same SuperGPQA split, sampled responses, and post-training recipe as O1-Pruner, but replace the response-length term in the target construction with measured generation energy. For each sampled trajectory, we record the GPU energy consumed during inference using the Intelligence per Watt instrumentation, assign that energy value to the corresponding question-response pair, and construct the pruning target by rewarding correctness while preferring lower-energy correct responses rather than shorter correct responses.

The energy values are normalized within the same target-building pipeline used for the length-based O1-Pruner targets, preserving the $\alpha = 5$, $K = 12$, and $k = 2$ setup. Thus, the energy-aware model differs from the O1-Pruner baseline only in the efficiency signal used during target construction. We then fine-tune Qwen3.5-27B with the same one-epoch, last-48-layer, 2×10^{-6} learning-rate,

effective-batch-size-32, 10,000-token configuration, and evaluate it on the same held-out SuperGPQA questions under the fixed grader and harnesses.

5 Results

We answer the three questions in order. Harness selection alone moves IPJ by up to an order of magnitude, energy is set by how much a configuration generates and at what precision, and a token-efficiency post-training method does not improve agent energy. We first report our measured numbers and then interpret the mechanisms behind them.

5.1 Quantitative Evaluation

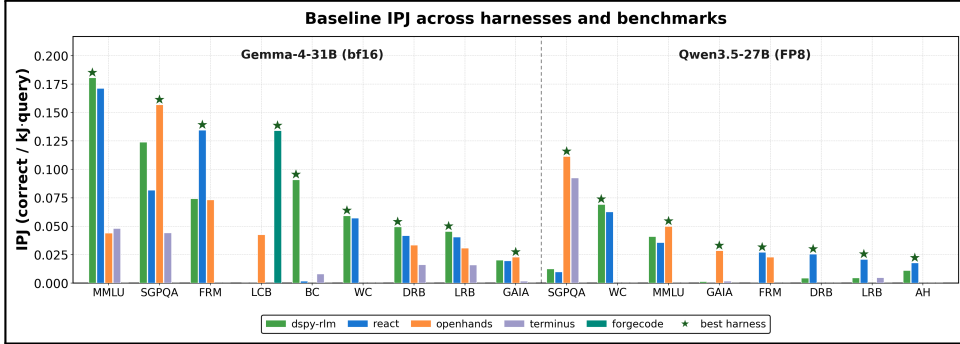


Figure 2: Per-query IPJ for every combination, with the star marking the most efficient harness.

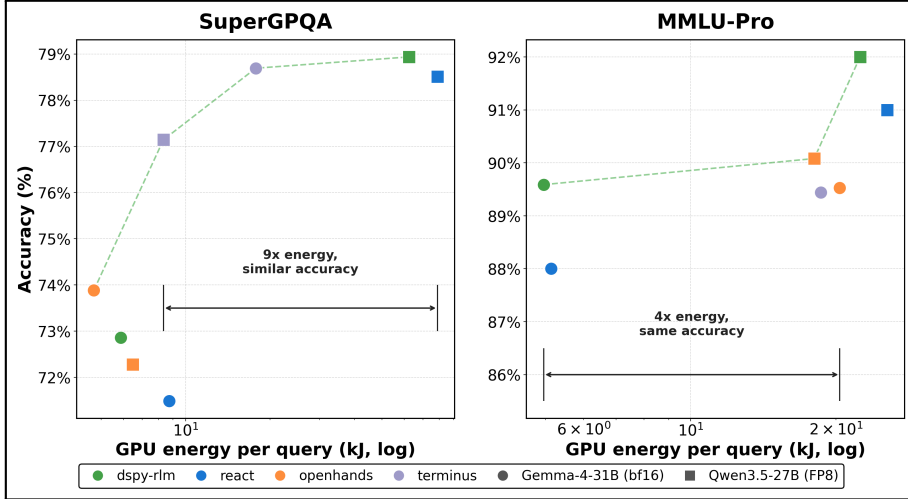


Figure 3: Accuracy versus GPU energy per query on SuperGPQA and MMLU-Pro. The dashed curve is the accuracy and energy Pareto frontier.

On a fixed model and benchmark the most and least efficient harness differ by up to $54\times$ in IPJ, with a median gap of $3\times$ (Figure 2). The most efficient harness is not fixed, shifting from dspey-rlm on most knowledge tasks to openhands or react on others, so no single scaffold is universally efficient. On SuperGPQA and MMLU-Pro, the lighter scaffold matches the heavier one within a few accuracy points at 4 to $9\times$ lower energy (Figure 3).

Per-query energy is governed by output volume. Across the 59 scorable cells, output tokens explain 94% of the variance in energy against 26% for turn count (Figure 4). The per-token rate is set by numerical precision, near 12 J in bf16 and 8 J in FP8, and generation dominates, with each output token costing roughly 10 J and each input token under 0.1 J. Energy is therefore decode-bound and precision-dependent rather than a function of turns or context length.

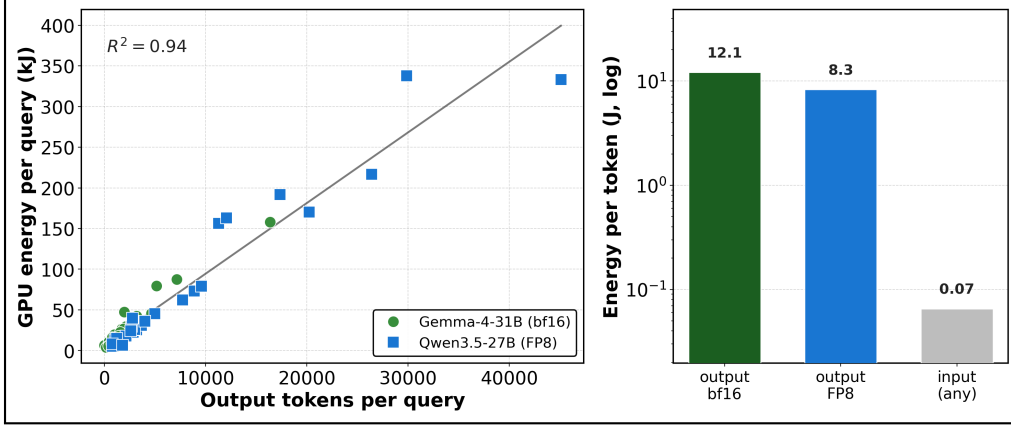


Figure 4: Left: GPU energy per query versus output tokens across all cells. Right: per-token energy by precision and for input versus output tokens.

The extra energy of heavier harnesses rarely buys accuracy. Comparing the lightest and heaviest harness per task across 18 cells, the heavier harness raises accuracy on only 4, leaves it unchanged on 6, and lowers it on 8 (Table 1). Agentic scaffolds pay off on genuinely multi-step tasks such as browsecomp and frames, and waste energy on knowledge question answering.

Regime	Cells	Acc. Δ	Examples
Energy buys accuracy	4	+0.02 to +0.49	browsecomp, frames
Strictly dominated	8	-0.02 to -0.90	livecodebench, deepresearchbench
Iso-accuracy, wasted	6	< 0.02	MMLU-Pro, WildChat

Table 1: Lightest versus heaviest harness per task, by accuracy change, across 18 model-task cells.

We evaluate post-training on SuperGPQA by comparing base Qwen, O1-Pruner, and the energy-aware O1 variant across ReAct, dspy-rlm, OpenHands, and Terminus. The token-only O1-Pruner lowers output tokens on ReAct and dspy-rlm, but its accuracy drops on every harness by 6 to 21 points under the fixed GPT grader (Figure 5, Table 2).

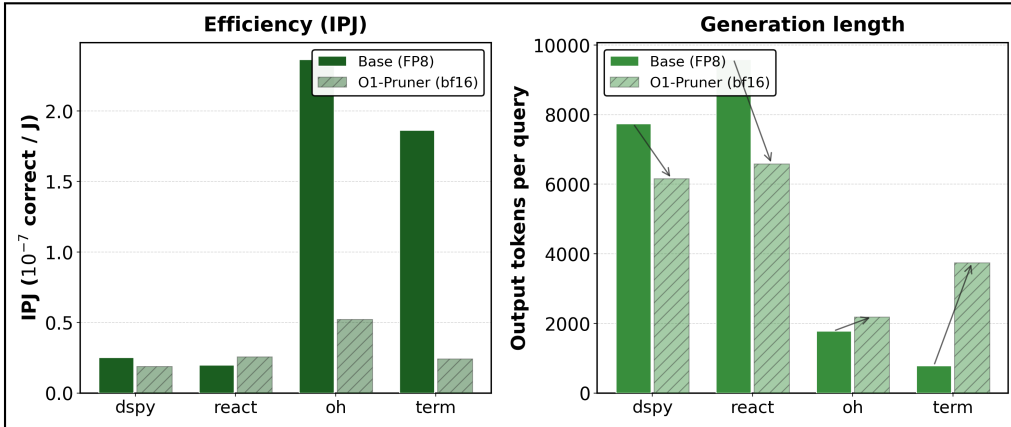


Figure 5: O1-Pruner versus base on SuperGPQA. Left: IPJ per harness. Right: output tokens per harness.

Because the base model is served in FP8 and the post-trained checkpoints are served in bf16, we also report an FP8-equivalent comparison using the same $1.5\times$ energy correction used above. After this correction, O1-Pruner has higher relative IPJ than base on ReAct, dspy-rlm, and Terminus, but remains below base on OpenHands (Figure 6).

Harness	Acc.	Output	kJ/query	IPJ
dspy-rlm, base	0.789	7735	62.4	0.25
dspy-rlm, pruned	0.582	6164	61.7	0.19
react, base	0.785	9588	78.9	0.20
react, pruned	0.727	6583	65.4	0.26
openhands, base	0.723	1785	6.5	2.37
openhands, pruned	0.576	2189	22.0	0.52
terminus, base	0.771	788	8.3	1.86
terminus, pruned	0.659	3748	54.3	0.24

Table 2: Base versus O1-Pruner on SuperGPQA under the GPT grader (gpt-5-nano). Output is mean output tokens per query, energy is kJ per query, and IPJ is in units of 10^{-7} correct per joule.

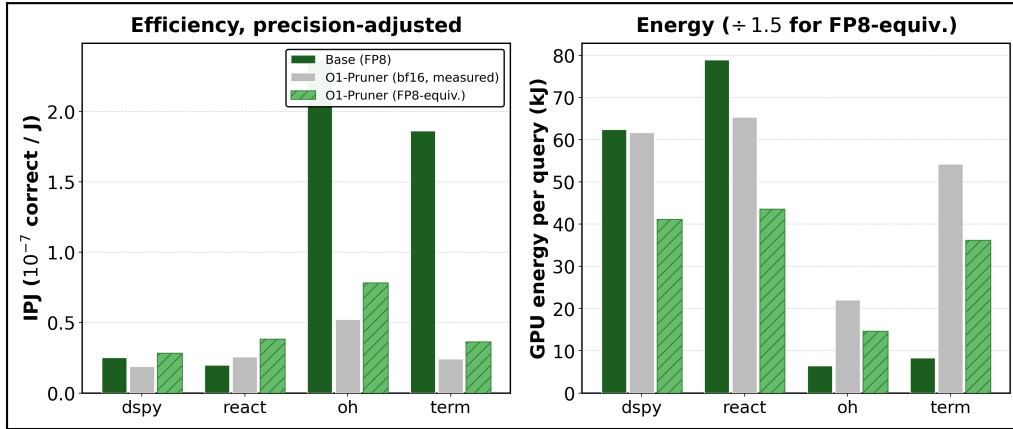


Figure 6: O1-Pruner with the bf16-to-FP8 precision penalty removed. Left: IPJ for base (FP8), the measured pruned model (bf16), and the pruned model rescaled to an FP8-equivalent. Right: GPU energy per query under the same three settings.

The energy-aware variant improves on token-only O1-Pruner in three of four harnesses on FP8-equivalent IPJ (Figure 7, Table 3). It also has higher accuracy than O1-Pruner on all four harnesses. On ReAct, FP8-equivalent relative IPJ rises from $1.63\times$ for O1-Pruner to $1.72\times$ for energy-aware O1. On Terminus, accuracy is 0.755 versus 0.743 for base Qwen, measured energy falls from 75.8 to 45.9 kJ/query, and FP8-equivalent relative IPJ reaches $2.51\times$. On OpenHands, both post-trained models remain below the base model.

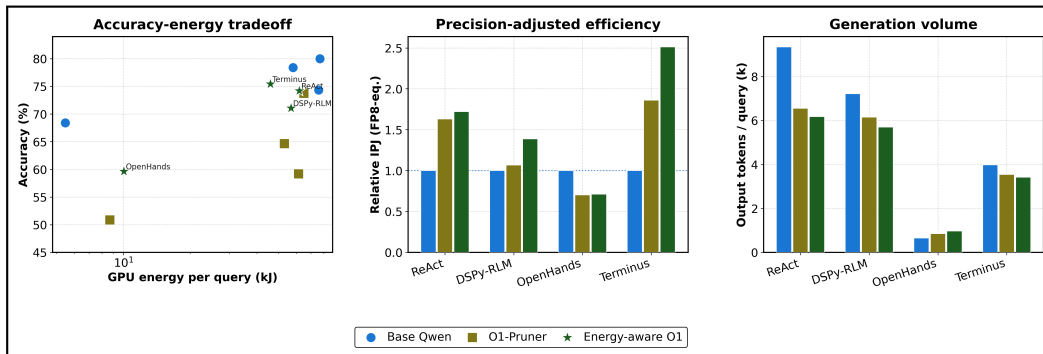


Figure 7: Energy-aware O1 versus token-only O1-Pruner and base Qwen on SuperGPQA. Left: accuracy versus measured GPU energy per query. Middle: FP8-equivalent IPJ normalized to the base model within each harness using the same $1.5\times$ bf16-to-FP8 correction as Figure 6. Right: output tokens per query.

Harness	Model	Acc.	Output	kJ/query	FP8-eq. Rel. IPJ
ReAct	base	0.800	9348	76.8	1.00
ReAct	O1-Pruner	0.737	6555	65.1	1.63
ReAct	energy-aware O1	0.742	6176	62.1	1.72
dspy-rlm	base	0.784	7220	58.1	1.00
dspy-rlm	O1-Pruner	0.592	6155	61.6	1.07
dspy-rlm	energy-aware O1	0.711	5699	56.9	1.39
OpenHands	base	0.684	653	5.5	1.00
OpenHands	O1-Pruner	0.509	856	8.7	0.70
OpenHands	energy-aware O1	0.596	972	10.1	0.71
Terminus	base	0.743	3984	75.8	1.00
Terminus	O1-Pruner	0.646	3546	53.1	1.86
Terminus	energy-aware O1	0.755	3418	45.9	2.51

Table 3: Base, token-only O1-Pruner, and energy-aware O1 on SuperGPQA. Output is mean output tokens per query. Relative IPJ normalizes each harness to its base Qwen run and applies the same $1.5\times$ bf16-to-FP8 correction to post-trained models.

6 Discussion

The results support IPJ as a first-class objective for agent inference. Agent cost is set by the full trajectory, so the harness is part of the optimization problem rather than an implementation detail. To our knowledge, this is the first work to motivate task-specific harness selection as a significant practical lever for agent energy efficiency. Across fixed model and benchmark pairs, changing the harness moved IPJ by as much as $54\times$, with a median $3\times$ gap. These gains require no new weights and no retraining; they come from matching the scaffold to the task. For factual and exam-style question answering, lightweight harnesses often preserve accuracy while avoiding repeated decoding. For genuinely multi-step tasks, heavier agent loops can be justified.

The post-training results demonstrate that naive token-only efficiency methods are not sufficient for agent inference. Tokens explain much of measured energy in aggregate, but a token-only objective does not know which scaffold will consume those tokens, how often the agent will re-enter the model, or whether a shorter trace will lower correctness. O1-Pruner shortened verbose ReAct-style trajectories but lengthened terse OpenHands and Terminus trajectories, and its accuracy dropped on all four harnesses. In this setting, length is a lossy proxy: it can reward policies that look efficient in a single response while wasting energy or losing accuracy over the full agent trajectory.

The energy- and agent-aware variant is more successful because it optimizes the quantity that deployment actually pays for: measured joules over complete traces. The method is not uniformly superior, but it is consistently stronger than token-only pruning on the post-training tradeoff: it improves the accuracy-energy frontier relative to O1-Pruner and achieves large FP8-equivalent IPJ gains on ReAct and Terminus. Harness selection and post-training should be designed together, so the policy is optimized on the energy profile induced by the scaffold it will actually use.

7 Conclusion

We studied agent inference energy as a trajectory-level objective using NVML telemetry across 67 model-task-harness configurations and a post-training study on Qwen3.5-27B. The results support three claims. First, task-specific harness selection can substantially improve IPJ, often preserving accuracy while cutting energy by 4 to $9\times$ and in some cases changing IPJ by up to $54\times$. Second, token-only pruning is not reliable for agent efficiency: O1-Pruner reduced measured IPJ on most harnesses and lost 6 to 21 accuracy points under the fixed grader. Third, replacing token length with measured energy produces a better post-training signal, with the energy-aware O1 variant outperforming token-only pruning and reaching $2.51\times$ relative IPJ on Terminus after the precision correction.

These results argue for treating deployment scaffolds and post-training objectives as a single optimization problem. A practical system should first select a harness suited to the task, then train or route policies using energy measured on that harness rather than token counts alone. Energy-aware

post-training is promising, and an exciting direction is the combination with harness selection: use the right scaffold for the task, then optimize the policy on the energy profile that scaffold actually induces.

8 Team Contributions

- **Ravenor Davion:** Infrastructure for the preliminary configuration runs. Implementation of the O1-Pruner post-training method and of the novel energy-aware method.
- **Cynthia Wang:** Evaluation and experiment analysis of baseline and O1-Pruner runs, as well as logic for the energy-aware method. Figures, plots, and literature review.

Changes from Proposal First, we made the novel method agent-trajectory-aware, so the energy-aware reward scores energy measured over the full agent trajectory rather than a single forward pass and reflects how the model is actually deployed (Section 3.2). Second, we were compute-bound. Presampling $K = 12$ references per question and fine-tuning a 27B model under per-query energy telemetry was expensive, so we ran post-training exclusively for the O1-Pruner method and our novel energy-aware method.

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